Pearson Edexcel Level 1/Level 2 – style practice paper

GCSE (9–1) Mathematics

1MA1/1H & 2H – Higher Tier

Topic 5: Probability

Centre Number		**Candidate Number**	
Surname		**Other names**	

Practice questions in the style of Pearson Edexcel GCSE Mathematics (1MA1). Every question is original; the layout, command words, mark allocations and mark scheme (M, A, B marks) follow the real exam format. Calculator rules are given in the instructions below.

Time allowed	About 100 minutes (roughly 1 minute per mark)
Instructions	• Use **black** ink or ball-point pen. • Answer **all** questions. • Answer the questions in the spaces provided – there may be more space than you need. • Calculators must **not** be used in Section A. Calculators may be used in Section B. • Diagrams are **NOT** accurately drawn, unless otherwise indicated. • You must **show all your working**. • **Unless stated, give non-exact answers to 3 significant figures.**
Information	• The total mark for this paper is **98**. • The marks for **each** question are shown in brackets – use this as a guide as to how much time to spend on each question. • Questions labelled with an asterisk (*) are ones where the quality of your written communication will be assessed.
Advice	• Read each question carefully before you start to answer it. • Keep an eye on the time. • Try to answer every question. • Check your answers if you have time at the end.

Name: _____ Date: _____ Mark: _____ / 98

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Section A: Non-calculator

You must not use a calculator for questions 1 to 12.

1 A bag contains 5 red counters, 3 blue counters and 2 green counters. A counter is taken at random.

(a) Write down the probability that the counter is blue. **(1)**

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(b) Work out the probability that the counter is **not** green. **(2)**

.....

(c) The counter is put back. This is done 150 times in total. Work out an estimate for the number of times a red counter is taken. **(2)**

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(Total for Question 1 is 5 marks)

2 A biased coin is thrown 200 times. The results are shown in the table.

Outcome	**Heads**	**Tails**
Frequency	124	76

(a) Write down an estimate for the probability that the coin lands on heads. **(2)**

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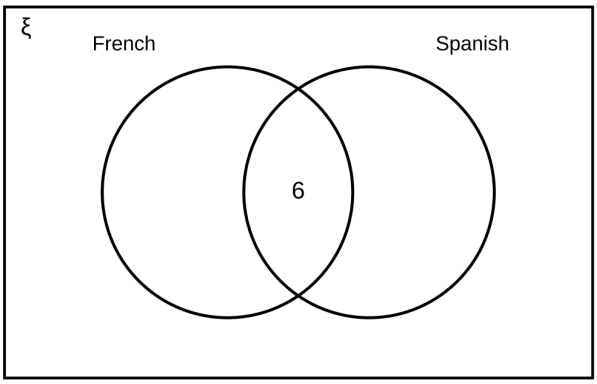
(b) The coin is thrown 500 more times. Estimate the number of tails. **(2)**

.....

(Total for Question 2 is 4 marks)

3 40 students were asked if they study French or Spanish. 22 study French, 15 study Spanish and 6 study both French and Spanish.

(a) Complete the Venn diagram. **(3)**



(b) One of the 40 students is chosen at random. Find the probability that the student studies French but not Spanish. **(1)**

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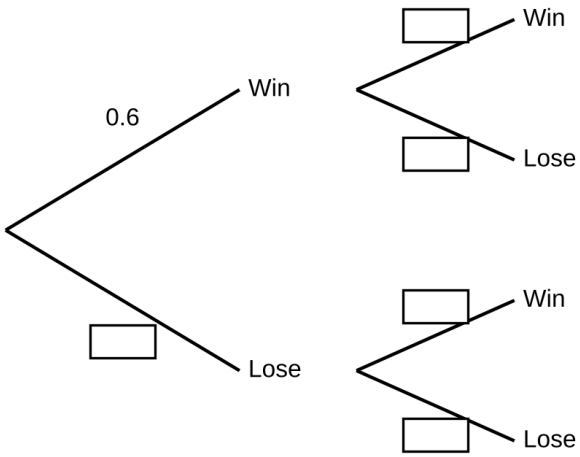
(c) A student is chosen at random from those who study Spanish. Find the probability that the student also studies French. **(2)**

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(Total for Question 3 is 6 marks)

4 Sam plays two games of chess. The probability that Sam wins any game is 0.6.

(a) Complete the probability tree diagram. **(3)**



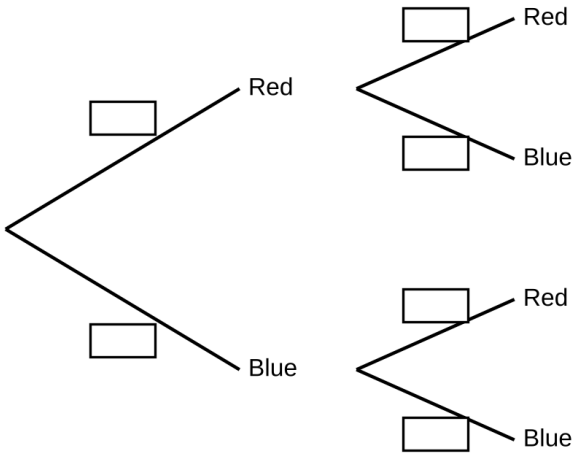
(b) Work out the probability that Sam wins exactly one of the two games. **(3)**

(c) Work out the probability that Sam wins at least one of the two games. **(2)**

(Total for Question 4 is 8 marks)

5 A bag contains 4 red counters and 6 blue counters. Two counters are taken at random **without replacement**.

(a) Complete the probability tree diagram. **(3)**



(b) Work out the probability that both counters are red. **(2)**

.....

(c) Work out the probability that the two counters are different colours. **(3)**

.....

(Total for Question 5 is 8 marks)

6 120 students were asked how they travel to school. The incomplete two-way table shows some of the results.

(a) Complete the two-way table. **(3)**

	Walk	**Bus**	**Car**	**Total**
Boys	24	30	16	
Girls	18			50
Total		52	26	120

(b) One of the 120 students is picked at random. Find the probability that the student is a girl who travels by bus. **(2)**

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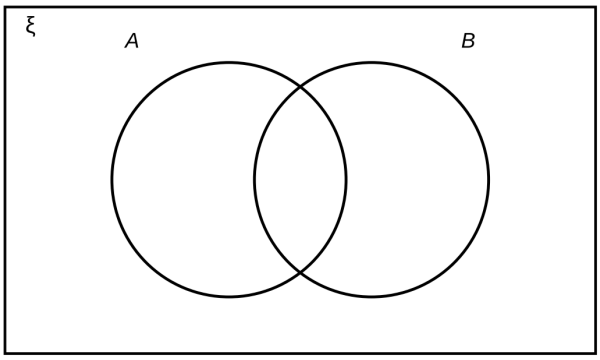
(c) A student who walks to school is picked at random. Work out the probability that the student is a boy. **(2)**

.....

(Total for Question 6 is 7 marks)

7 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ $A = \{\text{even numbers}\}$ $B = \{\text{multiples of } 3\}$

(a) Complete the Venn diagram by writing every number from ξ in the correct region. **(3)**



(b) A number is chosen at random from ξ. Find the probability that the number is in $A \cup B$. **(2)**

.....

(c) Find $P(A' \cap B)$. **(2)**

.....

(Total for Question 7 is 7 marks)

8 Events A and B are independent. $P(A) = 0.3$ and $P(B) = 0.5$.

(a) Work out $P(A \text{ and } B)$. **(2)**

.....

(b) Work out $P(A \text{ or } B)$. **(2)**

.....

(Total for Question 8 is 4 marks)

9 A fair six-sided dice is rolled twice and the two scores are added together.

(a) Complete the sample space table. **(2)**

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3						
4						
5						
6						

(b) Find the probability that the total score is 8. **(1)**

.....

(c) Find the probability that the total score is a prime number. **(2)**

.....

(Total for Question 9 is 5 marks)

10 A bag has 3 red counters and n blue counters. Two counters are taken at random without replacement. The probability that both counters are red is $\frac{1}{15}$.

(a) Show that $n^2 + 5n - 84 = 0$. **(4)**

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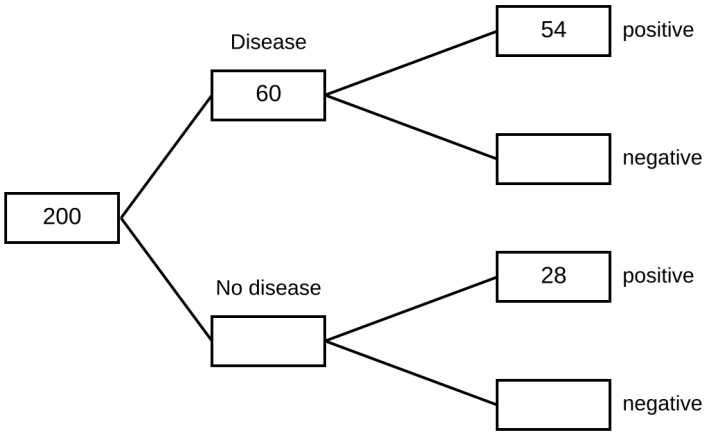
(b) Solve $n^2 + 5n - 84 = 0$ to find the number of blue counters. **(3)**

.....

(Total for Question 10 is 7 marks)

11 200 people took a medical test. 60 of them have the disease. Of the people who have the disease, 54 tested positive. Of the people who do not have the disease, 28 tested positive.

(a) Complete the frequency tree. **(3)**



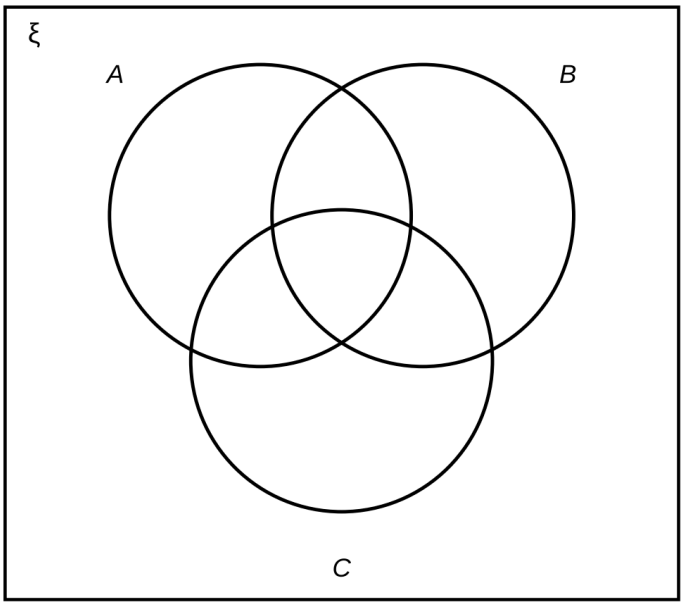
(b) One of the people who tested positive is chosen at random. Work out the probability that this person has the disease. **(2)**

.....

(Total for Question 11 is 5 marks)

12 60 people were asked which of three newspapers, A, B and C, they read. 25 read A, 20 read B and 18 read C. 8 read both A and B, 5 read both B and C, 6 read both A and C and 3 read all three.

(a) Complete the Venn diagram. **(4)**



(b) One of the 60 people is chosen at random. Find the probability that this person reads exactly two of the newspapers. **(2)**

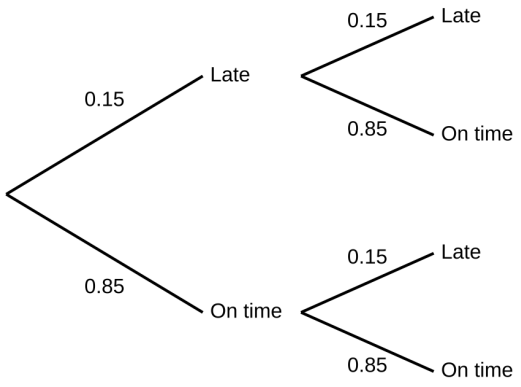
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(Total for Question 12 is 6 marks)

Section B: Calculator

A calculator may be used for questions 13 to 18.

13 The probability that a train is late is 0.15 on any day. Rohan catches the train on two days.



Work out the probability that the train is late on at least one of the two days. **(3)**

.....

(Total for Question 13 is 3 marks)

14 The probability that Hana is late for school on any day is 0.15. School is open on 3 days.

(a) Work out the probability that she is late on exactly one of the 3 days. **(3)**

.....

(b) Work out the probability that she is never late in the 3 days. **(2)**

.....

(Total for Question 14 is 5 marks)

15 A spinner has four sections: red, blue, green and yellow. The probability of red is x, blue is $2x$, green is 0.3 and yellow is 0.1.

(a) Work out the value of x. **(3)**

.....

(b) The spinner is spun 250 times. Work out the expected number of times it lands on blue. **(2)**

.....

(Total for Question 15 is 5 marks)

16 In a game, a player rolls a fair six-sided dice. If the score is 6 they win £5. If the score is 4 or 5 they win £1. Otherwise they win nothing. The game costs £1 to play.

Work out the expected profit (or loss) from one play. **(4)**

.....

(Total for Question 16 is 4 marks)

17 Two students are chosen at random from a class of 30 students (18 girls and 12 boys) without replacement.

(a) Work out the probability that they are both girls. Give your answer as a fraction in its simplest form. **(3)**

.....

(b) Work out the probability that there is one girl and one boy. **(3)**

.....

(Total for Question 17 is 6 marks)

18 Mia has a bag of 12 sweets. 5 are mint, 4 are toffee and 3 are fruit. She takes 3 sweets at random without replacement.

Work out the probability that all 3 sweets are mint. Give your answer correct to 3 significant figures. **(3)**

.....

(Total for Question 18 is 3 marks)

TOTAL FOR PAPER IS 98 MARKS

END

MARK SCHEME – Topic 5: Probability

Guidance on marking: M marks are method marks, awarded for a correct method (even if the final answer is wrong). **A** marks are accuracy marks, awarded only if the relevant M mark has been gained. **B** marks are independent marks (no method needed). **C** marks are communication marks. **P** marks are process marks (a correct approach to a problem). **ft** = follow through from a previous incorrect answer. **oe** = or equivalent. **cao** = correct answer only. **SC** = special case. If the answer on the answer line differs from the working, the answer line is marked unless the working is clearly the intended answer. Correct answers with no working score full marks unless the question says 'You must show your working'.

Question	Working	Answer	Mark	Notes
1(a)	B1 $\frac{3}{10}$ (oe)	$\frac{3}{10}$	1 AO1	
1(b)	M1 $1-\frac{2}{10}$ or $\frac{8}{10}$ A1 $\frac{4}{5}$ (oe)	$\frac{4}{5}$	2 AO1	
1(c)	M1 $\frac{5}{10}\times 150$ A1 75	75	2 AO3	
2(a)	M1 $\frac{124}{200}$ (relative frequency) A1 0.62 (oe)	0.62	2 AO2	
2(b)	M1 $500\times\frac{76}{200}$ or 500×0.38 A1 190	190	2 AO3	
3(a)	B1 French only $=16$ B1 Spanish only $=9$ B1 neither $=40-(16+6+9)=9$	16, 9, 9	3 AO3	
3(b)	B1 $\frac{16}{40}$ (oe, ft)	$\frac{2}{5}$	1 AO1	
3(c)	M1 denominator 15 or numerator 6 A1 $\frac{6}{15}=\frac{2}{5}$	$\frac{2}{5}$	2 AO3	
4(a)	B1 0.4 on the 'Lose' branch (first game) B1 0.6, 0.4 on the upper second-game branches B1 0.6, 0.4 on the lower second-game branches		3 AO1	
4(b)	M1 0.6×0.4 or 0.4×0.6 M1 $0.6\times 0.4+0.4\times 0.6$ A1 0.48	0.48	3 AO1	
4(c)	M1 $1-0.4\times 0.4$ or $0.36+0.24+0.24$ A1 0.84	0.84	2 AO3	
5(a)	B1 $\frac{4}{10}, \frac{6}{10}$ first branches B1 $\frac{3}{9}, \frac{6}{9}$ after red B1 $\frac{4}{9}, \frac{5}{9}$ after blue		3 AO1	
5(b)	M1 $\frac{4}{10}\times\frac{3}{9}$ A1 $\frac{2}{15}$ (oe, $\frac{12}{90}$)	$\frac{2}{15}$	2 AO1	

Question	Working	Answer	Mark	Notes
5(c)	M1 $\frac{4}{10} \times \frac{6}{9}$ or $\frac{6}{10} \times \frac{4}{9}$ M1 $\frac{24}{90} + \frac{24}{90}$ A1 $\frac{8}{15}$ (oe)	$\frac{8}{15}$	3 AO1	
6(a)	B1 Boys total $= 70$ (or Girls Car $= 10$) B1 two more correct entries B1 all correct: Boys: 24, 30, 16, 70; Girls: 18, 22, 10, 50; Total: 42, 52, 26, 120		3 AO3	
6(b)	M1 $\frac{22}{120}$ (ft) A1 $\frac{11}{60}$	$\frac{11}{60}$	2 AO1	
6(c)	M1 $\frac{24}{42}$ (ft) A1 $\frac{4}{7}$	$\frac{4}{7}$	2 AO3	
7(a)	B1 $A \cap B$: 6, 12 B1 A only: 2, 4, 8, 10 and B only: 3, 9 B1 outside both: 1, 5, 7, 11		3 AO1	
7(b)	M1 $\frac{8}{12}$ A1 $\frac{2}{3}$	$\frac{2}{3}$	2 AO1	
7(c)	M1 identifies $\{3, 9\}$ A1 $\frac{2}{12} = \frac{1}{6}$	$\frac{1}{6}$	2 AO2	
8(a)	M1 0.3×0.5 A1 0.15	0.15	2 AO1	
8(b)	M1 $0.3 + 0.5 - 0.15$ A1 0.65	0.65	2 AO1	
9(a)	B2 all correct (B1 for at least 8 correct): row 3: 4 to 9; row 4: 5 to 10; row 5: 6 to 11; row 6: 7 to 12		2 AO1	
9(b)	B1 $\frac{5}{36}$	$\frac{5}{36}$	1 AO1	
9(c)	M1 identifies totals 2, 3, 5, 7, 11 ($1 + 2 + 4 + 6 + 2 = 15$) A1 $\frac{15}{36} = \frac{5}{12}$	$\frac{5}{12}$	2 AO3	
10(a)	M1 $P(\text{red}) = \frac{3}{n+3}$ and $P(\text{second red}) = \frac{2}{n+2}$ M1 $\frac{3}{n+3} \times \frac{2}{n+2} = \frac{1}{15}$ M1 $90 = (n+3)(n+2)$ C1 $n^2 + 5n + 6 = 90 \Rightarrow n^2 + 5n - 84 = 0$		4 AO3	
10(b)	M1 $(n+12)(n-7) = 0$ A1 $n = 7$ and $n = -12$ A1 7 blue counters (reject -12)	7	3 AO3	
11(a)	B1 140 (no disease) B1 6 (disease, negative) B1 112 (no disease, negative)		3 AO3	

Question	Working	Answer	Mark	Notes
11(b)	M1 $54 + 28 = 82$ positive tests A1 $\frac{54}{82} = \frac{27}{41}$	$\frac{27}{41}$	2 AO3	
12(a)	B1 3 in the middle B1 $A \cap B$ only $= 5$, $B \cap C$ only $= 2$, $A \cap C$ only $= 3$ B1 A only $= 14$, B only $= 10$, C only $= 10$ B1 neither $= 13$	3; 5, 2, 3; 14, 10, 10; 13	4 AO3	
12(b)	M1 $5 + 2 + 3 = 10$ A1 $\frac{10}{60} = \frac{1}{6}$	$\frac{1}{6}$	2 AO3	
13	M1 $1 - 0.85 \times 0.85$ M1 $1 - 0.7225$ A1 0.2775	0.2775	3 AO3	
14(a)	M1 $0.15 \times 0.85 \times 0.85$ M1 $3 \times 0.15 \times 0.85^2$ A1 0.325 (0.325125)	0.325	3 AO3	
14(b)	M1 0.85^3 A1 0.614 (0.614125)	0.614	2 AO1	
15(a)	M1 $x + 2x + 0.3 + 0.1 = 1$ M1 $3x = 0.6$ A1 0.2	0.2	3 AO1	
15(b)	M1 250×0.4 A1 100	100	2 AO1	
16	M1 $\frac{1}{6} \times 5 = 0.83\ldots$ M1 $\frac{2}{6} \times 1 = 0.33\ldots$ M1 $\frac{7}{6} - 1$ (expected winnings minus cost) A1 $+£\frac{1}{6}$ (about 17p profit per play)	profit of about 17p	4 AO3	
17(a)	M1 $\frac{18}{30} \times \frac{17}{29}$ M1 $\frac{306}{870}$ A1 $\frac{51}{145}$	$\frac{51}{145}$	3 AO1	
17(b)	M1 $\frac{18}{30} \times \frac{12}{29}$ M1 $2 \times \frac{216}{870}$ A1 $\frac{72}{145}$	$\frac{72}{145}$	3 AO1	
18	M1 $\frac{5}{12} \times \frac{4}{11} \times \frac{3}{10}$ M1 $\frac{60}{1320}$ A1 0.0455	0.0455	3 AO3	

Where marks are lost in Topic 5

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Fractions of a total:** probabilities in one experiment add to 1. 'Not' = 1 − P.
- ☐ **Tree diagrams:** multiply along the branches, add between the branches. Write the product next to each outcome before you add.
- ☐ **Without replacement:** the denominator drops by 1 on the second pick and the numerator may change too.
- ☐ **'At least one':** use $1 - P(\text{none})$ — it saves lots of work.
- ☐ **Venn diagrams:** fill the intersection first, then work outwards. Check the regions add to the total in ξ.
- ☐ **Conditional probability:** the denominator is the number in the group you are told about ('given that…'), not the total.
- ☐ **Independent events:** $P(A \text{ and } B) = P(A) \times P(B)$. Mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$.
- ☐ **Expected number** = probability × number of trials. It is an estimate, not a guarantee.